

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

Name: Muhammad Usman Roll No: 23BSCYS006

Score: _____ Signature of the Lab Tutor: _____ Date: _____

Hands-On: Easy IP

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Objective:

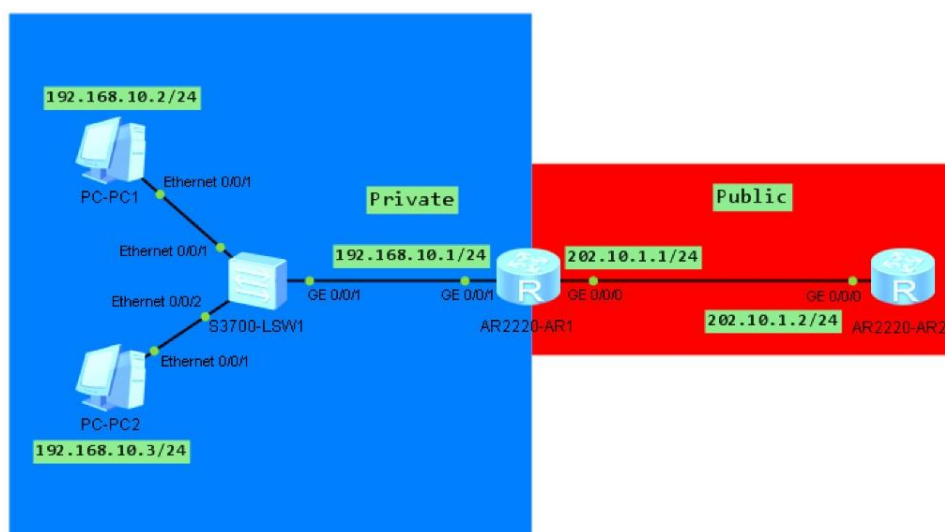
- Configure Easy IP on router

Learning Objectives:

As a result of this lab section, you should achieve the following tasks:

- Translation of addresses between networks (NAT)

Topology



**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

Scenario

As shown in Figure, the private network is shown to the left of the router, and the public network is shown to the right of the router. To enable a PC on the private network to communicate with a server on the Internet, we need to configure **Easy IP** on the router and map private addresses **192.168.0.X** (private) to a single public addresses **202.10.1.1** (public).

Easy IP

The Easy IP configuration is very similar in configuration to that of dynamic network address translation, relying on the creation of an access control list rule for defining the address range to which translation is to be performed and application of the **nat outbound** command. The main difference is in the absence of the **address group** command since no address pool is used in the configuration of Easy IP. Instead the outbound interface address is used to represent the external address, in this case that being external address 200.10.1.1 of interface GigabitEthernet0/0/0. Additionally it is necessary that port address translation be performed, and as such the **no-pat** parameter cannot be implemented where an address group does not exist. The **nat outbound 2000** represents a binding between the NAT operation and the access control list rule detailing the address range to which the translation will apply.

1. Run **system-view**
 1. The system view is displayed.
2. Configure outbound NAT.
 1. Configure Easy IP without an address pool.
 1. Run **interface** *interface-type interface-number* [*.subnumber*]
The interface view or sub-interface view is displayed.
 2. Run **nat outbound** *acl-number* [**interface** *interface-type interface-number* [*.subnumber*]] [*vrp vrrpid*]
3. Run the **display nat outbound** [**acl** *acl-number* | **address-group** *group-index* | **interface** *interface-type interface-number* [*.subnumber*]] command to verify the configuration of outbound NAT.

Tasks

Step 1 Preparing the environment

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname AR1
```

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
```

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

[Huawei] sysname AR2

Step 2 Configure Router AR1 and AR2

Configure the router AR1 interfaces to have IP addresses.

```
[AR1] interface GigabitEthernet 0/0/0
[AR1-GigabitEthernet0/0/0] ip address 202.10.1.1 24
[AR1-GigabitEthernet0/0/0] q
[AR1] interface GigabitEthernet 0/0/1
[AR1-GigabitEthernet0/0/1] ip address 192.168.10.1 24
[AR1-GigabitEthernet0/0/1] q
```

Configure the router AR2 interfaces to have IP addresses.

```
[AR2] interface GigabitEthernet 0/0/0
[AR2-GigabitEthernet0/0/0] ip address 202.10.1.2 24
[AR2-GigabitEthernet0/0/0] q
```

Step 3 Configure PCs

Configure IP address on PCs.

Host Name:	PC1		
MAC Address:	54-89-98-97-71-93		
IPv4 Configuration			
☒ Static	☐ DHCP	☐ Obtain DNS server address automatically	
IP Address:	192 , 168 , 10 , 2	DNS1:	0 , 0 , 0 , 0
Subnet Mask:	255 , 255 , 255 , 0	DNS2:	0 , 0 , 0 , 0
Gateway:	192 , 168 , 10 , 1		
IPv6 Configuration			
☒ Static	☐ DHCPv6		
IPv6 Address:	::		
Prefix Length:	128		
IPv6 Gateway:	::		

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

Host Name:	PC2		
MAC Address:	54-89-98-3F-0D-9C		
IPv4 Configuration			
☒ Static ☐ DHCP ☐ Obtain DNS server address automatically			
IP Address:	192 , 168 , 10 , 3	DNS1:	0 , 0 , 0 , 0
Subnet Mask:	255 , 255 , 255 , 0	DNS2:	0 , 0 , 0 , 0
Gateway:	192 , 168 , 10 , 1		
IPv6 Configuration			
☒ Static ☐ DHCPv6			
IPv6 Address:	::		
Prefix Length:	128		
IPv6 Gateway:	::		

Step 4 Verify the connectivity

```
PC> ping 202.10.1.2
```

```
Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break Request
timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!
--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

Step 5 Configure Easy IP on Router AR1

We need to configure to translate Private IP Addresses to one Public IP Address.

```
[AR1] acl 2000
[AR1-acl-basic-2000] rule permit source 192.168.10.0 0.0.0.255
[AR1-acl-basic-2000] q
[AR1] interface GigabitEthernet 0/0/0
```

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

```
[AR1-GigabitEthernet0/0/0] nat outbound 2000
[AR1-GigabitEthernet0/0/0] q
```

```
[AR1] display nat outbound NAT
Outbound Information:
```

```
-----
Interface                Acl      Address-group/IP/Interface      Type   ---
-----
GigabitEthernet0/0/0      2000          202.10.1.1      easyip
-----
Total : 1
```

The type listing of easyip makes it clearly understood when Easy IP has been successfully configured.

Step 6 Verify the connectivity

```
PC> ping 202.10.1.2
Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break
From 202.10.1.2: bytes=32 seq=1 ttl=254 time=94 ms From
202.10.1.2: bytes=32 seq=2 ttl=254 time=78 ms From
202.10.1.2: bytes=32 seq=3 ttl=254 time=47 ms From
202.10.1.2: bytes=32 seq=4 ttl=254 time=31 ms From
202.10.1.2: bytes=32 seq=5 ttl=254 time=16 ms
```

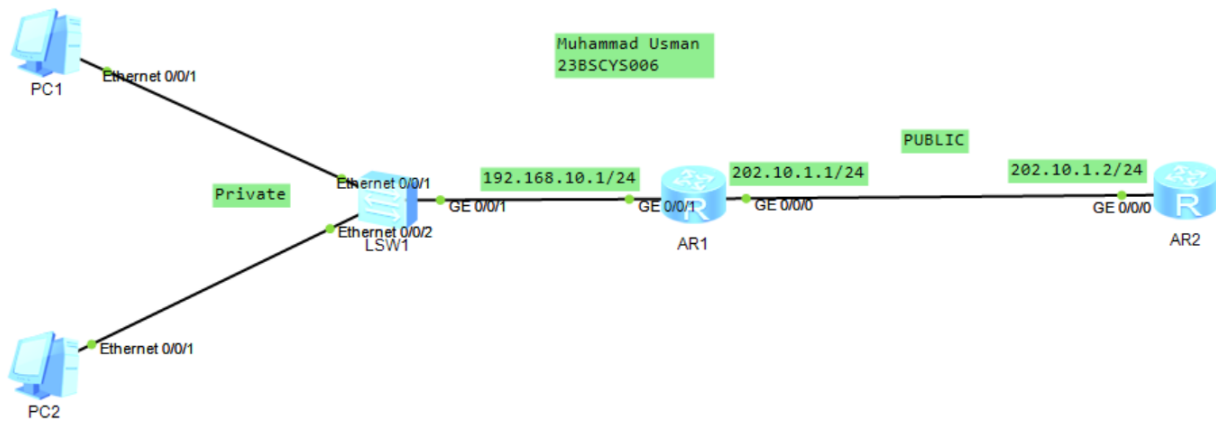
```
--- 202.10.1.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
0.00% packet loss
   round-trip min/avg/max = 16/53/94 ms
PC>
```

Tasks

Try a different topology and demonstrate your learning.

Upload the `.topo` `.efz` `.cfg` files

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**



```
PC1
Basic Config  Command  MCPacket  UdpPacket  Console
From 202.10.1.2: bytes=32 seq=3 ttl=254 time=16 ms
From 202.10.1.2: bytes=32 seq=4 ttl=254 time=62 ms
From 202.10.1.2: bytes=32 seq=5 ttl=254 time=47 ms
--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 16/40/62 ms
PC>ping 202.10.1.2
Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break
From 202.10.1.2: bytes=32 seq=1 ttl=254 time=47 ms
From 202.10.1.2: bytes=32 seq=2 ttl=254 time=31 ms
From 202.10.1.2: bytes=32 seq=3 ttl=254 time=31 ms
From 202.10.1.2: bytes=32 seq=4 ttl=254 time=31 ms
From 202.10.1.2: bytes=32 seq=5 ttl=254 time=47 ms
--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/37/47 ms
PC>
```

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

```
AR1
AR1  AR2
Protocol      : ICMP(1)
SrcAddr Vpn   : 192.168.10.254
DestAddr Vpn  : 202.10.1.2
Type Code IcmpId : 0  8  60061
NAT-Info
  New SrcAddr  : 202.10.1.1
  New DestAddr : ----
  New IcmpId   : 10241

Protocol      : ICMP(1)
SrcAddr Vpn   : 192.168.10.254
DestAddr Vpn  : 202.10.1.2
Type Code IcmpId : 0  8  60062
NAT-Info
  New SrcAddr  : 202.10.1.1
  New DestAddr : ----
  New IcmpId   : 10242

Protocol      : ICMP(1)
SrcAddr Vpn   : 192.168.10.254
DestAddr Vpn  : 202.10.1.2
Type Code IcmpId : 0  8  60063
NAT-Info
  New SrcAddr  : 202.10.1.1
  New DestAddr : ----
  New IcmpId   : 10243

Protocol      : ICMP(1)
SrcAddr Vpn   : 192.168.10.254
DestAddr Vpn  : 202.10.1.2
Type Code IcmpId : 0  8  60064
NAT-Info
  New SrcAddr  : 202.10.1.1
  New DestAddr : ----
  New IcmpId   : 10244

Total : 5
[23BSCYS006-AR-1]
```

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING BACHELOR OF
SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 14**

```
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 1
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 1

Interface                IP Address/Mask      Physical  Protocol
GigabitEthernet0/0/0     202.10.1.1/24        up        up
GigabitEthernet0/0/1     192.168.10.1/24      up        up
GigabitEthernet0/0/2     unassigned            down      down
NULL0                    unassigned            up        up(s)
[23BSCYS006-AR-1]
```